

Hyper Hamiltonian Generalized Petersen Graphs

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Abstract

Assume that n and k are positive integers with $n \geq 2k + 1$. A non-hamiltonian graph G is *hypo hamiltonian* if $G - v$ is hamiltonian for any $v \in V(G)$. It is proved that the generalized Petersen graph $P(n, k)$ is *hypo hamiltonian* if and only if $k = 2$ and $n \equiv 5 \pmod{6}$. Similarly, a hamiltonian graph G is *hyper hamiltonian* if $G - v$ is hamiltonian for any $v \in V(G)$. In this paper, we will give some necessary conditions and some sufficient conditions about the hyper hamiltonian generalized Petersen graphs. In particular, $P(n, k)$ is not hyper hamiltonian if n is even and k is odd. We also prove that $P(3k, k)$ is hyper hamiltonian if and only if k is odd. Moreover, $P(n, 3)$ is hyper hamiltonian if and only if n is odd and $P(n, 4)$ is hyper hamiltonian if and only if $n \neq 12$. Furthermore, $P(n, k)$ is hyper hamiltonian if k is even with $k \geq 6$ and $n \geq 2k + 2 + (4k - 1)(4k + 1)$, and $P(n, k)$ is hyper hamiltonian if $k \geq 5$ is odd and n is odd with $n \geq 6k - 3 + 2k(6k - 2)$.

1 Definitions and notations

For the graph definitions and notations we follow [5]. $G = (V, E)$ is a *graph* if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that $V(G)$ is the *vertex set* and $E(G)$ is the *edge set* of G . We delimited a path P in a graph from a vertex v_0 to v_n by $\langle v_0, v_1, v_2, \dots, v_n \rangle$. A cycle is a path with at least three vertices such that its first vertex is the same as the last vertex. A cycle is a *hamiltonian cycle*

if it traverses every vertex of G exactly once. A graph is *hamiltonian* if it has a hamiltonian cycle.

Assume that n and k are positive integers with $n \geq 2k + 1$. We use $\oplus$ to denote addition in integer modular n , Z_n . The *generalized Petersen graph* $P(n, k)$ is the graph with vertex set $\{i \mid 0 \leq i \leq n - 1\} \cup \{i' \mid 0 \leq i \leq n - 1\}$ and edge set $\{(i, i \oplus 1) \mid 0 \leq i \leq n - 1\} \cup \{(i, i') \mid 0 \leq i \leq n - 1\} \cup \{(i', (i \oplus k)') \mid 0 \leq i \leq n - 1\}$. It is proved that $P(n, k)$ is not hamiltonian if and only if $k = 2$ and $n \equiv 5 \pmod{6}$ [2, 4, 6, 8]. However, all non-hamiltonian generalized Petersen graphs satisfy another interesting property. A non-hamiltonian graph G such that $G - v$ is hamiltonian for any $v \in V(G)$ is called a *hypo hamiltonian* graph. It is proved that the set of hypo hamiltonian generalized Petersen graphs is actually the set of non-hamiltonian generalized Petersen graphs [6].

Similarly, a hamiltonian graph G is *hyper hamiltonian* if $G - v$ is hamiltonian for any $v \in V(G)$. We are interested in the recognition of hyper hamiltonian generalized Petersen graphs. In [1], it is proved that $P(n, 1)$ is hyper hamiltonian if and only if n is odd and $P(n, 2)$ is hyper hamiltonian if and only if $n \equiv 1, 3 \pmod{6}$.

In this paper, we will give some necessary conditions and some sufficient conditions about the hyper hamiltonian generalized Petersen graphs. In particular, $P(n, k)$ is not hyper hamiltonian if n is even and k is odd. We also proved that $P(3k, k)$ is hyper hamiltonian if and only if k is odd. Moreover, $P(n, 3)$ is hyper hamiltonian if and only if n is odd; $P(n, 4)$ is hyper hamiltonian if and only if $n \neq 12$. Furthermore, $P(n, k)$ is hyper hamiltonian if k is even with $k \geq 6$ and $n \geq 2k + 2 + (4k - 1)(4k + 1)$, and $P(n, k)$ is hyper hamiltonian if $k \geq 5$ is odd with $n \geq 6k - 3 + 2k(6k - 2)$.

In the following section, we give some prelimi-

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naries about hyper hamiltonian graphs. In Section 3, we prove that $P(3k, k)$ is hyper hamiltonian if and only if k is odd. In Section 4, we prove that $P(n, 3)$ is hyper hamiltonian if and only if n is odd. In Section 5, we prove that $P(n, 4)$ is hyper hamiltonian if and only if $n \neq 12$. In Section 6, we prove that $P(n, k)$ is hyper hamiltonian if k is even with $k \geq 6$ and $n \geq 2k + 2 + (4k - 1)(4k + 1)$, and $P(n, k)$ is hyper hamiltonian if k is odd with $k \geq 5$ and n is odd with $n \geq 6k - 3 + 2k(6k - 2)$. In the final section, we give a concluding remark.

2 Preliminaries

Lemma 1 *Any bipartite graph is not hyper hamiltonian.*

Proof: Assume that G is a hyper bipartite graph with n vertices. Then G has a cycle of length n and $n - 1$. This is impossible because G has no cycle of odd length. Hence, the lemma is proved.

By the definition of the generalized Petersen graphs, the next lemma is obtained.

Lemma 2 *$P(n, k)$ is bipartite if and only if n is even and k is odd. Thus, $P(n, k)$ is not hyper hamiltonian if n is even and k is odd.*

With Lemma 2, we may ask if $P(n, k)$ is not hyper hamiltonian if and only if n is even and k is odd. However, the statement is not true because of the following theorem.

Theorem 1 [1] *$P(n, 1)$ is hyper hamiltonian if and only if n is odd and $P(n, 2)$ is hyper hamiltonian if and only if $n \equiv 1, 3 \pmod{6}$.*

Thus, $P(n, 2)$ with n is even is neither bipartite nor hyper hamiltonian. However, we desire to know if there are other generalized Petersen graphs that are not hyper hamiltonian.

Let us consider the generalized Petersen graph $P(12, 4)$ shown in Figure 1(a). Suppose that $P(12, 4)$ is hyper hamiltonian. Then $P(12, 4) - 1$ is hamiltonian. We note that the vertex set $\{0', 4', 8'\}$ induces a complete graph K_3 . Any hamiltonian cycle in $P(12, 4) - 1$ must traverse the vertex set $\{0', 4', 8'\}$ consecutively because $P(12, 4)$ is a cubic graph. Similarly, any hamiltonian cycle traverses the vertex sets $\{1', 5', 9'\}$, $\{2', 6', 10'\}$, and $\{3', 7', 11'\}$ consecutively. For this reason, we can define a graph $P'(12, 4)$ obtained from $P(12, 4)$ by shrinking the vertices $0', 4',$ and $8'$ into a vertex $< 0 >$, shrinking the

vertices $1', 5',$ and $9'$ into a vertex $< 1 >$, shrinking the vertices $2', 6',$ and $10'$ into a vertex $< 2 >$ and shrinking the vertices $3', 7',$ and $11'$ into a vertex $< 3 >$ as shown in Figure 1(b). Thus, $P(12, 4) - 1$ is hamiltonian if and only if $P'(12, 4) - 1$ is hamiltonian. However, $P'(12, 4)$ is a bipartite graph with 8 vertices in each partite sets. Therefore, $P'(12, 4) - 1$ is not hamiltonian. We get a contradiction. Thus, $P(12, 4) - 1$ is not hamiltonian.

Alspach et al. [3] proposed an interesting model, called *lattice model*, to describe a hamiltonian cycle of the generalized Petersen graphs. With the lattice model, a *lattice diagram* for a generalized Petersen graphs is a labeled graph in the (x, y) -plane that possesses a closed or an open Eulerian trail. By appropriately interpreting the edges in the diagram, the Eulerian trail corresponds to a hamiltonian cycle of a generalized Petersen graph.

A lattice diagram $D(13, 4)$ for a generalized Petersen $P(13, 4)$, for example, is shown in Figure 2. In this diagram, there is an Eulerian trail $\langle 0, 1, 5, 9, 10, 6, 2, 3, 7, 11, 12, 8, 7, 6, 5, 4, 0 \rangle$. By appropriately interpreting the edges in the diagram, it corresponds to a hamiltonian cycle $\langle 0, 1, 1', 5', 9', 9, 10, 10', 6', 2', 2, 3, 3', 7', 11', 11, 12, 12', 8', 8, 7, 6, 5, 4, 4', 0', 0 \rangle$ in $P(13, 4)$.

In [3], Alspach et al. introduced this scheme and proved that $P(n, k)$ is hamiltonian for all $n \geq (4k - 1)(4k + 1)$ if k is even and for all $n \geq (2k - 1)(3k + 1)$ if k is an odd integer with $k \geq 3$.

The lattice model is described as follows. For more details, see [3]. In lattice model, a *lattice graph* L consists of lattice points in the (x, y) -plane. Two lattice points (a_1, b_1) and (a_2, b_2) in L are said to be adjacent if and only if $|a_1 - a_2| + |b_1 - b_2| = 1$. Hence, the degree of each vertex is less than or equal to 4. Let n and k be positive integers with $n \geq 2k + 1$. A *labeled lattice graph* $L(n, k)$ is obtained from lattice graph L by labeling the lattice points following the rule: if a lattice point (a, b) is labeled with an integer i with $0 \leq i \leq n - 1$, then $(a + 1, b)$ is labeled with $i \oplus 1$ and $(a, b - 1)$ is labeled with $i \oplus k$. A lat-

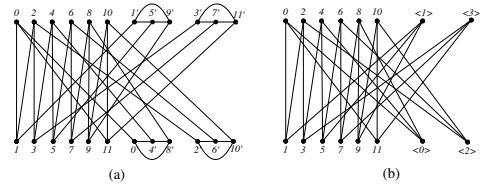


Figure 1: (a) $P(12, 4)$ and (b) $P'(12, 4)$.

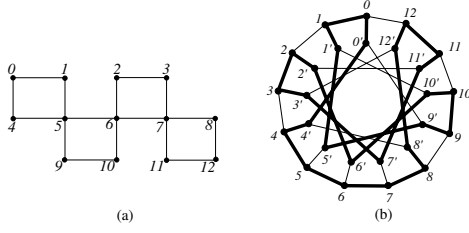


Figure 2: (a) A $D(13, 4)$ and (b) its corresponding hamiltonian cycle in $P(13, 4)$.

tice diagram for $P(n, k)$, denoted as $D(n, k)$, is a subgraph of $L(n, k)$ induced by the vertices with labels $0, 1, \dots, n-1$ such that it possesses either a closed or an open Eulerian trail. A traversal of the Eulerian trail in $D(n, k)$ obeys the following rules:

1. The trail does not change the direction when it passes through a vertex of degree 4.
2. Each label $0, 1, \dots, n-1$ is encountered by the traversal once in a vertical direction and once in a horizontal direction.
3. If $D(n, k)$ has an open Eulerian trail, then the two vertices of odd degree must have the same label and not of both being of degree 3.

The correspondence between an Eulerian trail of a lattice diagram $D(n, k)$ and a hamiltonian cycle of the generalized Petersen graph $P(n, k)$ is built by interpreting the edges in $L(n, k)$. The interpretations of edges in $L(n, k)$ are given as follows:

1. The vertical edge $(i, i \oplus k)$ in $L(n, k)$ corresponds to an edge $(i', (i \oplus k)')$ in $P(n, k)$.
2. The horizontal edge $(i, i \oplus 1)$ in $L(n, k)$ corresponds to an edge $(i, i \oplus 1)$ in $P(n, k)$.
3. Two edges in different directions incident to the vertex i of degree 2 in $L(n, k)$ correspond to an edge (i, i') in $P(n, k)$.

It is not difficult to see that an Eulerian trail in $L(n, k)$ corresponds to a hamiltonian cycle of $P(n, k)$ and any hamiltonian cycle of $P(n, k)$ can be converted into an Eulerian trail in $D(n, k)$. Hence, finding a hamiltonian cycle of a generalized Petersen graph $P(n, k)$ is finding an appropriate lattice diagram for $P(n, k)$ [3].

In addition, Alspach et al. [3] proposed an amalgamating mechanism to generate lattice diagrams with various sizes. For example, we have a lattice

diagram $D(4k, k)$ in Figure 3(a) and a lattice diagram $D(4k+2, k)$ in Figure 3(b). By identifying the vertex with label $(4k-1)$ of $D(4k, k)$ and the vertex with label 0 of $D(4k+2, k)$ and relabeling all vertices of $D(4k+2, k)$ by adding $(4k-1)$ to all the labels, we obtain a lattice diagram $D(8k+1, k)$ shown in Figure 3(c). Thus, $P(8k+1, k)$ is hamiltonian. With this mechanism, we can amalgamate r copies of $D(4k, k)$ and s copies of $D(4k+2, k)$, where $r \geq 1$ and $s \geq 0$, to make a lattice diagram $D(4k + (r-1)(4k-1) + s(4k+1), k)$. Hence, $P(4k + (r-1)(4k-1) + s(4k+1), k)$ is hamiltonian for $r \geq 1$ and $s \geq 0$.

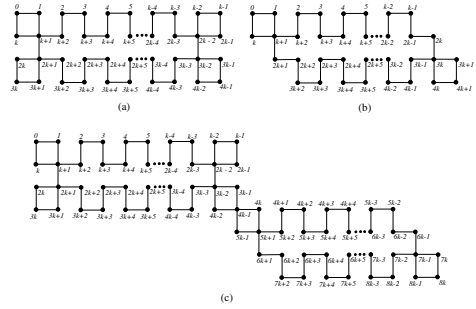
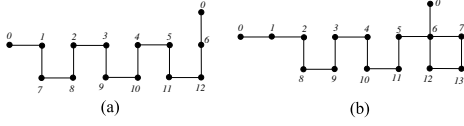


Figure 3: (a) A $D(4k, k)$, (b) a $D(4k+2, k)$, and (c) a $D(8k+1, k)$.

Now, we slightly modify the lattice diagram proposed in [3] to show that a generalized Petersen graph is hyper hamiltonian.

First, we give examples to demonstrate the variation. In Figure 4(a), an open Eulerian trail $\langle 0, 1, 7, 8, 2, 3, 9, 10, 4, 5, 11, 12, 6, 0 \rangle$ corresponds to the hamiltonian cycle $\langle 0, 1, 1', 7', 7, 8, 8', 2', 2, 3, 3', 9', 9, 10, 10', 4', 4, 5, 5', 11', 11, 12, 12', 6', 0', 0 \rangle$ of $P(13, 6) - 6$ using the same interpretation described above. The reason that the open Eulerian trail corresponds to the hamiltonian cycle of $P(13, 6) - 6$ is that the vertex with label 6 is traversed only in a vertical direction and all the other vertices are traversed in both directions. Again, the open Eulerian trail in Figure 4(b), $\langle 0, 1, 2, 8, 9, 3, 4, 10, 11, 5, 6, 7, 13, 12, 6, 0 \rangle$ corresponds to the hamiltonian cycle $\langle 0, 1, 2, 2', 8', 8, 9, 9', 3', 3, 4, 4', 10', 10, 11, 11', 5', 5, 6, 7, 7', 13', 13, 12, 12', 6', 0', 0 \rangle$ in $P(14, 6) - 1'$. In this diagram, the vertex with label 1 is traversed only in a horizontal direction and all the other vertices are traversed in both directions.

In other words, finding a lattice diagram $D_i^v(n, k)$ with an Eulerian trail such that the vertex with label i is traversed only in a vertical direction and all the other vertices are traversed in


 Figure 4: (a) A $D_6^v(13, 6)$ and (b) a $D_1^h(14, 6)$.

both directions is finding a hamiltonian cycle of $P(n, k) - i$. Similarly, finding a lattice diagram $D_i^h(n, k)$ with an Eulerian trail such that the vertex with label i is traversed only in a horizontal direction and all the other vertices are traversed in both directions is finding a hamiltonian cycle of $P(n, k) - i'$. Based on the fact that the automorphism group of $P(n, k)$ has at most two orbits $0, 1, 2, \dots, n-1$ and $0', 1', 2', \dots, (n-1)'$ [7], we need to construct a lattice diagram $D_i^v(n, k)$ and a lattice diagram $D_j^h(n, k)$ for some i and j with $0 \leq i, j \leq n-1$ to prove that $P(n, k)$ is hyper hamiltonian.

3 The generalized Petersen graph $P(3k, k)$

In this section, we prove that $P(3k, k)$ is hyper hamiltonian if and only if k is odd. We first prove that $P(3k, k)$ is not hyper hamiltonian if k is even. The proof is similar to the proof of that fact that $P(12, 4)$ is not hyper hamiltonian in Section 2.

Let k be an even integer. Suppose that $P(3k, k)$ is hyper hamiltonian. Then, there exists a hamiltonian cycle of $P(3k, k) - 1$. By the definition of the generalized Petersen graph, the set $\{i', (i+k)', (i+2k)'\}$ induces a cycle of length 3 for $0 \leq i \leq k-1$. Therefore, any hamiltonian cycle in $P(3k, k) - 1$ will traverse the vertex set $\{i', (i+k)', (i+2k)'\}$ consecutively for $0 \leq i \leq k-1$. Therefore, vertices $i', (i+k)'$, and $(i+2k)'$ can be regarded as a vertex for $0 \leq i \leq k-1$. Let $P'(3k, k)$ be the graph obtained from $P(3k, k)$ by shrinking the vertices $i', (i+k)'$, and $(i+2k)'$ into a new vertex, say $[i]$, for $0 \leq i \leq k-1$. It is not difficult to see that $P(3k, k) - 1$ is hamiltonian if and only if $P'(3k, k) - 1$ is hamiltonian.

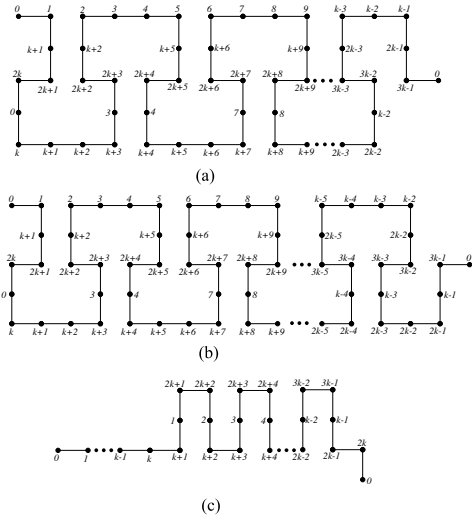
Now, we claim that $P'(3k, k)$ is a bipartite graph. Let $X = \{1, 3, 5, \dots, (3k-1)\} \cup \{[i] \mid 0 \leq i \leq k-1 \text{ and } i \text{ is even}\}$ and $Y = \{0, 2, 4, 6, \dots, (3k-2)\} \cup \{[i] \mid 0 \leq i \leq k-1 \text{ and } i \text{ is odd}\}$. It is easy to check that (X, Y) forms a bipartition of $P(3k, k)$ with $|X| = |Y| = 2k$. Hence, $P'(3k, k)$ is a bipartite graph. Thus, it has no cy-

cles of odd lengths. This implies that $P'(3k, k) - 1$ is not hamiltonian, and hence, $P(3k, k) - 1$ is not hamiltonian. Therefore, $P(3k, k)$ is not hyper hamiltonian if k is even.

Thus, we consider k is odd. Suppose $n = 1$. Obviously, $\langle 1, 1', 0', 2', 2, 1 \rangle$ forms a hamiltonian cycle for $P(3, 1) - 0$ and $\langle 0, 1, 1', 2', 2, 0 \rangle$ forms a hamiltonian cycle for $P(3, 1) - 0'$. Thus, $P(3, 1)$ is hyper hamiltonian.

Suppose $k = 3$. Then $\langle 1, 2, 3, 3', 0', 6', 6, 7, 8, 8', 2', 5', 5, 4, 4', 7', 1', 1 \rangle$ forms a hamiltonian cycle of $P(9, 3) - 0$ and $\langle 0, 1, 2, 3, 3', 6', 6, 7, 7', 1', 4', 4, 5, 5', 2', 8', 8, 0 \rangle$ forms a hamiltonian cycle for $P(9, 3) - 0'$. Thus, $P(9, 3)$ is hyper hamiltonian.

Suppose $k \geq 5$. A lattice diagram $D_{2k-1}^v(3k, k)$ with $k \equiv 1 \pmod{4}$ is shown in Figure 5(a) and a lattice diagram $D_{k-1}^v(3k, k)$ with $k \equiv 3 \pmod{4}$ is shown in Figure 5(b). Moreover, a lattice diagram $D_k^h(3k, k)$ is shown in Figure 5(c). Thus, $P(3k, k)$ is hyper hamiltonian.


 Figure 5: (a) A $D_{2k-1}^v(3k, k)$ with $k \equiv 1 \pmod{4}$, (b) a $D_{k-1}^v(3k, k)$ with $k \equiv 3 \pmod{4}$, and (c) a $D_k^h(3k, k)$ for odd k with $k \geq 5$.

Thus, we have the following theorem.

Theorem 2 $P(3k, k)$ is hyper hamiltonian if and only if k is odd.

4 The generalized Petersen graph $P(n, 3)$

Theorem 3 The generalized Peterson graph $P(n, 3)$ is hyper hamiltonian if and only if n is

odd.

Proof: By the definition of $P(n, k)$, we consider $n \geq 7$. By Lemma 2, $P(n, 3)$ is not hyper hamiltonian if n is even. Thus, we consider the case n is odd.

Suppose $n = 7$. Then $\langle 0, 0', 4', 1', 1, 2, 3, 3', 6', 2', 5', 5, 6, 0 \rangle$ is a hamiltonian cycle for $P(7, 3) - 4$ and $\langle 0, 1, 2, 3, 4, 4', 0', 3', 6', 2', 5', 5, 6, 0 \rangle$ is a hamiltonian cycle for $P(7, 3) - 1'$. Thus, $P(7, 3)$ is hyper hamiltonian.

Suppose $n = 9$. By Theorem 2, $P(9, 3)$ is hyper hamiltonian in Theorem 2.

Suppose $n \geq 11$. A lattice diagram $D_4^v(n, 3)$ is shown in Figure 6(a) and a lattice diagram $D_1^h(n, 3)$ is shown in Figure 6(b). Thus, $P(n, 3)$ is hyper hamiltonian.

Therefore, $P(n, 3)$ is hyper hamiltonian if and only if n is odd.

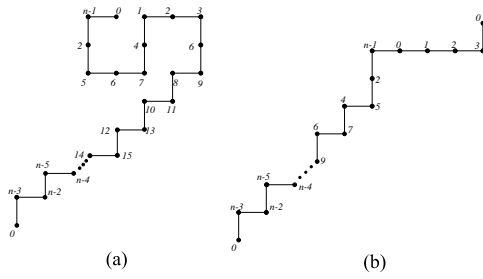


Figure 6: (a) A $D_4^v(n, 3)$ and (b) a $D_1^h(n, 3)$ for odd n with $n \geq 11$.

5 The generalized Petersen graph $P(n, 4)$

Theorem 4 *The generalized Petersen graph $P(n, 4)$ is hyper hamiltonian if and only if $n \neq 12$.*

Proof: By the definition of $P(n, k)$, we consider $n \geq 9$. With Theorem 2, $P(12, 4)$ is not hyper hamiltonian. Thus, we consider the cases $n \geq 9$ and $n \neq 12$.

Suppose $n = 9$. Then $\langle 0, 1, 2, 2', 7', 7, 6, 6', 1', 5', 5, 4, 3, 3', 8', 4', 0' \rangle$ is a hamiltonian cycle for $P(9, 4) - 8$ and $\langle 0, 1, 1', 5', 0', 4', 8', 3', 7', 2', 2, 3, 4, 5, 6, 7, 8, 0 \rangle$ is a hamiltonian cycle for $P(9, 4) - 6'$. Thus, $P(9, 4)$ is hyper hamiltonian.

Suppose $n = 10$. Then $\langle 0, 1, 2, 3, 3', 9', 5', 1', 7', 7, 8, 8', 2', 6', 6, 5, 4, 4', 0' \rangle$

is a hamiltonian cycle for $P(10, 4) - 9$ and $\langle 0, 1, 1', 5', 5, 4, 4', 8', 2', 2, 3, 3', 9', 9, 8, 7, 6, 6', 0', 0 \rangle$ is a hamiltonian cycle for $P(10, 4) - 7'$. Thus, $P(10, 4)$ is hyper hamiltonian.

Suppose $n = 11$. Then $\langle 0, 1, 1', 8', 8, 9, 9', 5', 5, 6, 7, 7', 3', 10', 6', 2', 2, 3, 4, 4', 0', 0 \rangle$ is a hamiltonian cycle for $P(11, 4) - 10$ and $\langle 0, 1, 1', 5', 9', 2', 2, 3, 3', 10', 6', 6, 5, 4, 4', 0', 7', 7, 8, 9, 10, 0 \rangle$ is a hamiltonian cycle for $P(11, 4) - 8'$. Thus, $P(11, 4)$ is hyper hamiltonian.

Suppose $n \geq 13$. In Figure 7(a), (b), (c), and (d), we have four lattice diagrams $D_{n-1}^v(n, 4)$ depending on the value of $n \pmod{4}$. In Figure 7(e), (f), (g), and (h), we have four lattice diagrams $D_{n-3}^h(n, 4)$ depending on the value of $n \pmod{4}$. Thus, $P(n, 4)$ is hyper hamiltonian.

Thus, $P(n, 4)$ is hyper hamiltonian if and only if $n \neq 12$.

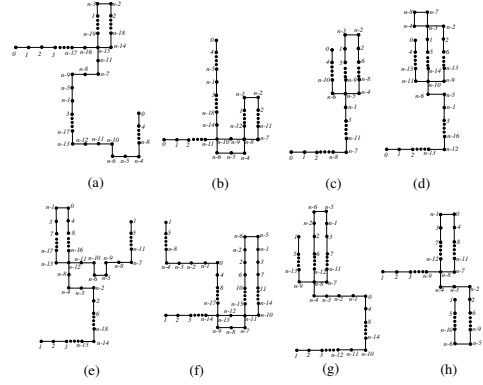


Figure 7: (a) A $D_{n-1}^v(n, 4)$ with $n \equiv 0 \pmod{4}$, (b) a $D_{n-1}^v(n, 4)$ with $n \equiv 1 \pmod{4}$, (c) a $D_{n-1}^v(n, 4)$ with $n \equiv 2 \pmod{4}$, (d) a $D_{n-1}^v(n, 4)$ with $n \equiv 3 \pmod{4}$, (e) a $D_{n-3}^h(n, 4)$ with $n \equiv 0 \pmod{4}$, (f) a $D_{n-3}^h(n, 4)$ with $n \equiv 1 \pmod{4}$, (g) a $D_{n-3}^h(n, 4)$ with $n \equiv 2 \pmod{4}$, and (h) a $D_{n-3}^h(n, 4)$ with $n \equiv 3 \pmod{4}$.

6 The generalized Petersen graph $P(n, k)$ with $k \geq 5$

With Theorems 1, 3, and 4, we can recognize those hyper hamiltonian Petersen graphs $P(n, k)$ with $k \leq 4$. Now, we consider the case $k \geq 5$. By Lemma 2, $P(n, k)$ is not hyper hamiltonian if n is even and k is odd. In this section, we will prove that $P(n, k)$ is hyper hamiltonian if (1) $k \geq 5$ is odd and n is odd with n sufficiently large with respect to k ; and (2) k is even with $k \geq 6$ and n is

sufficient large with respect to k . We will use the amalgamating mechanism, proposed by Alspach et al. [3], described in Section 2 to obtain suitable lattice diagrams.

We first consider the case that k is even with $k \geq 6$. We will use four basic lattice diagrams. Let $D(4k, k)$ and $D(4k+2, k)$ be the lattice diagrams shown in Figures 3(a) and 3(b), respectively. Moreover, let $D_k^v(2k+1, k)$ be the lattice diagram shown in Figure 8(a) and $D_1^h(2k+2, k)$ be the lattice diagram shown in Figure 8(b). For illustration, we amalgamate a $D_6^v(13, 6)$ and a $D(26, 6)$ to obtain a $D_6^v(38, 6)$ in Figure 8(c). Note that $38 = 13 + 26 - 1$ because a vertex is duplicated during the amalgamating mechanism. Similarly, we amalgamate a $D_1^h(14, 6)$ and a $D(24, 6)$ to obtain a $D_1^h(37, 6)$ in Figure 8(d).

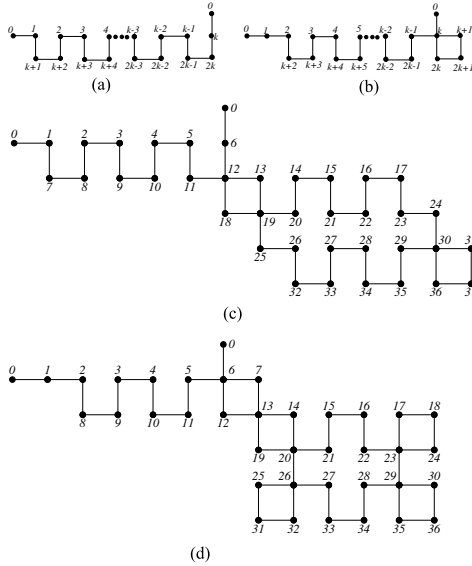


Figure 8: (a) $D_k^v(2k+1, k)$, (b) $D_1^h(2k+2, k)$, (c) $D_6^v(38, 6)$, and (d) $D_1^h(37, 6)$.

Assume that $n \geq 2k+2+(4k-1)(4k+1)$. Since $\gcd(4k-1, 4k+1) = 1$, there exist nonnegative integers r and s such that $n = 2k+1+r(4k-1)+s(4k+1)$. Similarly, there exist nonnegative integers t and u such that $n = 2k+2+t(4k-1)+u(4k+1)$. We can amalgamate one copy of $D_k^v(2k+1, k)$, r copies of $D(4k, k)$, and s copies of $D(4k+2, k)$ to make a lattice diagram $D_k^v(n, k)$. Thus, $P(n, k) - k$ is hamiltonian. Again, we can amalgamate one copy of $D_1^h(2k+2, k)$, t copies of $D(4k, k)$, and u copies of $D(4k+2, k)$ to make a lattice diagram $D_1^h(n, k)$. Thus, $P(n, k) - 1'$ is hamiltonian.

Thus, we have the following theorem.

Theorem 5 Suppose that k is even with $k \geq 6$. Then $P(n, k)$ is hyper hamiltonian if $n \geq 2k+2+(4k-1)(4k+1)$.

Now, we consider both k and n to be odd integers with $k \geq 5$. We will use four basic lattice diagrams. Let $D_0^v(6k-3, k)$ be the lattice diagram shown in Figure 9(a), $D_1^h(2k+1, k)$ be the lattice diagram shown in Figure 9(b), $D(2k, k)$ be the lattice diagram in Figure 9(c), and $D(6k-2, k)$ be the lattice diagram in Figure 9(d). For illustration, we amalgamate a $D_0^h(27, 5)$ and a $D(10, 5)$ to obtain a $D_0^h(37, 5)$ in Figure 9(e). We observe that $37 = 27 + 10$ which is different from the previous case. The two vertices labeled with 0 in $D_0^h(27, 5)$ are the terminal vertices of the open Eulerian trail. We identify one of the vertex labeled with 0 in $D_0^h(27, 5)$ with one of the vertex labeled with 0 in $D(10, 5)$ to obtain a $D_0^h(37, 5)$. Similarly, we amalgamate a $D_1^h(11, 5)$ and a $D(28, 5)$ to obtain a $D_1^h(39, 5)$ in Figure 8(f).

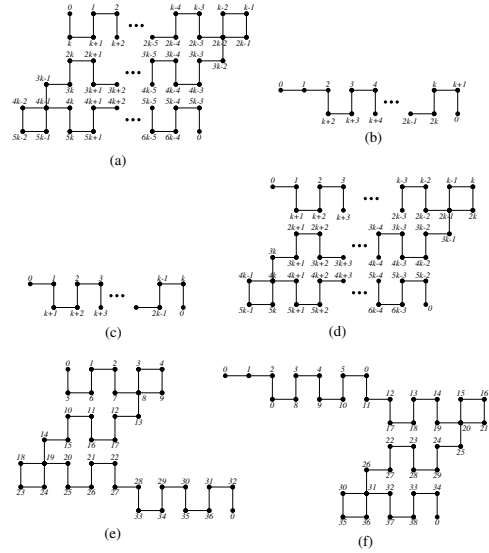


Figure 9: (a) $D_0^v(6k-3, k)$, (b) $D_1^h(2k+1, k)$, (c) $D(2k, k)$, (d) $D(6k-2, k)$, (e) $D_0^h(27, 5)$, and (f) $D_1^h(39, 5)$.

Assume that $n \geq 6k-3+(2k)(6k-2)$ and n is odd. Since $\gcd(2k, 6k-2) = 2$, there exist nonnegative integers r and s such that $n = 6k-3+r(2k)+s(6k-2)$. Moreover, there exist nonnegative integers t and u such that $n = 6k-3+t(2k)+u(6k-2)$. We can amalgamate one copy of $D_0^v(6k-3, k)$, r copies of $D(2k, k)$, and s copies of $D(6k-2, k)$ to make a lattice diagram $D_0^v(n, k)$. Similarly, we can amalgamate one copy of $D_1^h(2k+1, k)$, t copies of

$D(2k, k)$, and u copies of $D(6k - 2, k)$ to make a lattice diagram $D_1^h(n, k)$. Thus, $P(n, k)$ is hyper hamiltonian.

Theorem 6 Assume that k and n are odd integers with $k \geq 5$. Then $P(n, k)$ is hyper hamiltonian if $n \geq 6k - 3 + (2k)(6k - 2)$.

7 Concluding remark

In [1], it is proved that $P(n, 1)$ is hyper hamiltonian if and only if n is odd and $P(n, 2)$ is hyper hamiltonian if and only if $n \equiv 1, 3 \pmod{6}$. In [6], it is proved that $P(n, 2)$ is hypo hamiltonian if and only if $n \equiv 5 \pmod{6}$. In this paper, we proved that $P(3k, k)$ is hyper hamiltonian if and only if k is odd. Moreover, $P(n, 3)$ is hyper hamiltonian if and only if n is odd and $P(n, 4)$ is hyper hamiltonian if and only if $n \neq 12$. Furthermore, $P(n, k)$ is hyper hamiltonian if $n \geq 2k + 2 + (4k - 1)(4k + 1)$ and k is even with $k \geq 6$; $P(n, k)$ is hyper hamiltonian if k is odd with $k \geq 5$ and n is odd with $n \geq 6k - 3 + 2k(6k - 2)$. However, we conjecture that all non-bipartite generalized Petersen graphs except for $\{P(3k, k) \mid k \text{ is even}\} \cup \{P(n, 2) \mid n \equiv 0, 2, 4, 5 \pmod{6}\}$ are hyper hamiltonian.

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